

RIDDER, Kazakhstan

WMO: 362080

Lat: 50.333N

Lon: 83.550E

Elev: 811

StdP: 91.95

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.5	-28.6	-35.0	0.2	-32.4	-31.2	0.2	-28.3	16.7	-0.3	14.4	-0.5	1.2	90	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	28.7	17.7	27.0	17.1	25.4	16.4	19.2	26.1	18.1	25.0	17.2	23.9	2.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	13.2	22.7	15.5	12.1	21.6	14.4	11.3	20.6	58.3	26.1	54.6	24.9	51.6	23.9	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	8.6	6.3	DB	-35.3	31.1	4.5	1.7	-38.5	32.3	-41.2	33.3	-43.7	34.3	-47.0	35.6	(4)
(5)			WB	-35.4	21.4	4.5	1.6	-38.6	22.5	-41.2	23.5	-43.8	24.3	-47.0	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-13.0	-10.6	-4.0	4.3	10.6	15.8	17.2	15.8	10.1	3.9	-4.9	-10.9	(6)	
	DBStd	12.09	7.33	6.92	6.48	5.43	4.44	3.64	2.99	3.41	4.59	5.26	7.15	6.85	(7)	
	HDD10.0	3297	712	576	433	178	48	4	0	2	54	197	446	647	(8)	
	HDD18.3	5665	970	809	691	421	241	88	56	93	247	447	696	905	(9)	
	CDD10.0	726	0	0	0	8	67	179	224	181	58	8	1	0	(10)	
	CDD18.3	51	0	0	0	0	1	13	22	13	2	0	0	0	(11)	
	CDH23.3	884	0	0	0	1	47	227	286	264	58	1	0	0	(12)	
	CDH26.7	177	0	0	0	0	9	34	62	60	12	0	0	0	(13)	
(14) Wind	WSAvg	2.0	2.0	2.0	2.1	2.2	2.4	1.6	1.5	1.6	1.9	2.2	2.2	2.2	(14)	
(15) Precipitation	PrecAvg	639	19	21	34	58	76	76	88	65	53	61	53	35	(15)	
	PrecMax	824	36	38	64	142	191	153	160	146	149	119	137	75	(16)	
	PrecMin	395	3	7	11	11	20	21	39	1	9	20	22	6	(17)	
	PrecStd	113	10	9	13	32	40	34	33	34	33	25	26	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	6.3	13.6	22.5	27.8	29.4	31.0	31.0	28.3	21.6	14.1	6.4	(19)	
		MCWB	1.1	1.3	5.9	11.2	15.2	18.2	19.8	18.4	16.1	11.6	7.4	1.2	(20)	
	2%	DB	1.6	3.5	10.7	19.3	24.2	27.6	28.5	28.5	24.8	18.6	10.2	3.0	(21)	
		MCWB	-1.9	-0.4	4.3	9.7	13.8	17.2	18.3	17.3	14.2	10.2	4.7	-0.5	(22)	
	5%	DB	-1.2	1.3	8.1	16.7	22.1	26.0	26.5	26.3	22.2	15.5	7.1	0.1	(23)	
		MCWB	-3.6	-1.8	3.0	8.6	12.7	16.3	17.8	16.6	13.1	8.5	3.3	-2.3	(24)	
	10%	DB	-3.5	-1.0	5.5	14.2	19.8	24.2	24.7	24.4	19.6	12.3	4.4	-1.9	(25)	
		MCWB	-5.4	-3.3	1.4	7.1	11.5	15.8	17.2	16.0	11.9	6.9	1.5	-3.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	1.8	6.4	12.1	17.1	20.1	21.3	20.2	16.7	12.6	7.9	1.8	(27)	
		MCDB	4.4	5.3	12.0	20.2	25.1	26.8	27.7	26.9	26.6	19.8	12.9	5.9	(28)	
	2%	WB	-1.6	-0.1	4.8	10.4	14.7	18.5	19.7	18.2	14.9	10.5	5.2	-0.2	(29)	
		MCDB	1.3	3.1	9.7	18.4	22.7	25.5	26.3	26.3	22.9	17.6	9.4	2.6	(30)	
	5%	WB	-3.5	-1.6	3.2	9.0	13.5	17.3	18.5	17.2	13.6	9.0	3.4	-2.1	(31)	
		MCDB	-1.3	1.0	7.6	15.7	20.4	24.2	24.8	24.9	21.2	14.6	6.8	-0.2	(32)	
	10%	WB	-5.3	-3.2	1.7	7.6	12.2	16.4	17.5	16.3	12.2	7.4	1.5	-3.8	(33)	
		MCDB	-3.6	-0.9	5.1	13.3	18.5	22.9	23.5	23.3	18.9	12.0	4.4	-2.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	12.0	12.0	11.9	12.8	12.6	11.6	13.3	13.4	10.7	9.5	9.5	(35)	
		MCDBR	10.9	12.2	13.9	17.0	17.0	15.9	15.2	17.1	18.5	16.9	12.0	10.6	(36)	
	5% WB	MCWBR	8.3	8.9	8.9	9.6	9.1	8.0	7.8	8.6	9.7	9.7	7.7	7.9	(37)	
		MCWBR	10.5	11.4	12.6	15.8	15.1	14.0	13.4	15.3	17.1	15.8	11.6	10.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.279	0.311	0.362	0.418	0.403	0.401	0.423	0.388	0.350	0.345	0.310	0.282	(40)	
		taud	2.067	2.012	1.927	1.984	2.173	2.255	2.205	2.285	2.347	2.218	2.081	2.014	(41)	
	Ebn at Noon		725	796	817	817	853	857	829	842	839	756	671	642	(42)	
		Edh at Noon	83	115	152	162	140	130	135	119	100	94	82	76	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.45	2.55	3.88	4.61	5.80	6.28	5.81	5.39	3.93	2.12	1.22	1.03	(44)	
	RadStd		0.10	0.12	0.36	0.43	0.48	0.55	0.46	0.37	0.37	0.29	0.13	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page